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Experiment name: Data Analysis And Visualisation

Data Preprocessing And Visualization

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('student_performance.csv')

print(df.head())

print(df.isnull().sum())

df['Marks'] = df['Marks'].fillna(df['Marks'].mean())
df['Attendance'] = df['Attendance'].fillna(df['Attendance'].mean())

print(df.describe())

summary = df.groupby('Department')[['Marks', 'Attendance']].mean()
print(summary)

plt.bar(summary.index, summary['Marks'])
plt.xlabel('Department')
plt.ylabel('Average Marks')
plt.title('Average Marks by Department')
plt.show()

plt.plot(df['Student_ID'], df['Marks'], marker='o')
plt.xlabel('Student ID')
plt.ylabel('Marks')
plt.title('Student Marks')
plt.show()

pivot = df.pivot_table(
    values='Marks',
    index='Department',
    columns='Subject',
    aggfunc='mean',
    fill_value=0
)
print(pivot)

correlation = df.corr(numeric_only=True)
print(correlation)

sns.heatmap(correlation, annot=True)
plt.title('Correlation Matrix')
plt.savefig('bar_grapg.png')
plt.show()
```

